TAEUK IM

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INTERESTS

Satellite Networks, Orbital Edge Computing, Space Data Centers, Satellite Constellation Design, Mission Design for Cubesats

EDUCATION

M.S. in Aerospace Engineering *Sep 2025 - Aug 2027 (Expected)*
KAIST (Korea Advanced Institute of Science and Technology), Daejeon, South Korea *GPA 3.66 / 4.3 (3.67 / 4.0)*

- Advisor : Jihwan Choi ; ACAI Laboratory (Aerospace Communications & Applied Intelligence Lab, https://acai-kaist.github.io)

B.S. in Astronomy *Mar 2023 - Aug 2025*
Yonsei University, Seoul, South Korea *GPA 3.88 / 4.3 (3.72 / 4.0)*

- HIGH HONORS, 1ST SEMESTER, 2024 ; HONORS, 2ND SEMESTER, 2023

Undergraduate studies in Electrical Engineering (prior to transfer) *Mar 2018 - Dec 2020*
Sejong University, Seoul, South Korea *GPA 4.38 / 4.5 (3.95 / 4.0)*

CONFERENCES/PRESENTATIONS

T. Im, S. Kim, I. Choi, and J. Choi, "Performance Evaluation of Space Data Centers: DDSSO-Assisted LEO Mega-Constellations," Korean Institute of Information and Communications (KICS) Summer Conference 2026, Jun. 2026.

T. Im and J. Choi, "Multi-Shell Space Data Centers in the Sun-Synchronous Orbit-Assisted LEO Mega-Constellation," Korean Institute of Information and Communications (KICS) Winter Conference 2026, Feb. 2026.

PUBLICATIONS

1. S. Kim, T. Im, and J. P. Choi, "Integrated Design of Scheduling, Beamforming, and Reliable NOMA Transmission in LEO SATCOM," submitted to IEEE Transactions on Aerospace and Electronic Systems, 2026.
2. I. Choi, S. Kim, T. Im, and J. P. Choi, "Dynamic Virtual Network Embedding for Online Task Offloading in LEO Mega-Constellations," submitted to IEEE Transactions on Mobile Computing, 2026.

RESEARCH EXPERIENCE

Full-Time Master's Researcher *Sep 2025 - Present*
KAIST, Daejeon, Korea

- Assisted in background research by reviewing relevant academic papers and patents for the preparation of a government-funded basic research project proposal.
- Conducted a literature review on computation offloading in Space-Air-Ground Integrated Network (SAGIN), focusing on architectures, optimization methods, and DRL approaches.
- Proposed a multi-shell space data center architecture in an Sun Synchronous Orbit (SSO)-assisted LEO mega-constellation and have been conducting follow-up research on joint energy–latency optimization of toy problems using PPO and KKT-based approaches.

Undergraduate Research Assistant : Navigation, Control & Design for Autonomous Systems (NCDAS) Lab *Jul 2024 - May 2025*
Yonsei University, Seoul, Korea

- Developed a numerical two-body orbit propagator in MATLAB using ODE45. Verified conserved orbital quantities and cross-checked propagated states against an independent Keplerian propagator and GMAT's RK89 integrator across LEO, GEO, and elliptical orbit cases; extended the framework to include J2 perturbation.
- Developed an orbit-design framework to maximize the J2-induced RAAN precession rate for faster global coverage, combining MATLAB optimization with polynomial fitting to reduce repeated computation.
- Designed sun-synchronous and Molniya orbits, validating SSO nodal precession against theory and reproducing published long-term Molniya orbital-element evolution.

Undergraduate Research Assistant (Advisor : Jae Kyu Suhr) *Sep 2019 - Feb 2020*
Sejong University, Seoul, Korea

- Studied fundamentals of digital image processing, including edges, gradients, filters and transforms

- Learned core concepts of computer vision, including least squares, RANSAC and Viola-Jones object detection
- Studied basics of deep neural networks (DNNs), including CNN-based image classification and YOLO-based object detection

PROJECTS

Conceptual Design of a Reusable Unmanned Space Vehicle *Sep 2024 - Dec 2024*

- Defined an Earth-observation mission concept for a reusable unmanned space vehicle and led the GNC subsystem, designing the mission sequence from direct LEO insertion and on-orbit imaging through fuel-efficient deorbit, atmospheric reentry, and runway recovery for reuse.
- Designed and simulated a circular sun-synchronous repeat-ground-track operational orbit in MATLAB and GMAT for repeated daytime Earth observation, incorporating J2-induced nodal regression; analyzed altitude and inclination trades and verified multi-day ground-track repeatability.
- Coordinated with subsystem leads to translate payload, power, communications, propulsion, and structural constraints into the operational orbit, orbit maintenance concept, navigation architecture, and deorbit plan.

KOREA CANSAT COMPETITION *Jul 2024*

- Led the design of a vision-based CanSat mission to estimate the Sun's position from camera-derived elevation and azimuth, transforming observer-centered coordinates into Earth-centered inertial coordinates; extended the concept to general object identification and coordinate estimation

Literature Review on Computation Offloading in SAGIN: Architecture, Optimization, and DRL Approaches *Sep 2025*

- Analyzed SAGIN architectures and optimization problems across multiple studies
- Examined optimization formulations including task offloading, association control, resource allocation, and trajectory design
- Studied Lyapunov optimization for problem reformulation, MINLP solution methods (e.g., GBD and SCA with KKT-based convex subproblems), and DRL-based approaches (e.g., DDPG, DQN)

Computational Astronomy (Implemented the algorithm using Python) *Mar 2024 - Jun 2024*

- One-semester course project for "Computational Astronomy"
- Interpolation Methods : Linear interpolation, polynomial interpolation, cubic spline interpolation
- Optimization : χ^2 minimization and line minimization, Downhill Simplex (Amoeba) and Powell methods, Gauss–Newton and Levenberg–Marquardt algorithms
- Solver for differential Equations : Euler, Midpoint (Leapfrog), 4th-order Runge–Kutta
- Root-Finding Algorithms : Bisection, Secant, False-Position, Newton–Raphson methods, nonlinear system solving using Broyden's method

SKILLS

Technical: Python, MATLAB, GMAT, C++, Arduino, Github, Linux, LaTeX, Machine learning, Deep Reinforcement Learning, Implementation of optimization methods

WORK EXPERIENCE

Teaching Assistant

KAIST, Daejeon, Korea

- KAIST-Saudi Space Professional Development Program - KAIST Space Institute (Summer, 2026; Developed MATLAB modules for Walker Delta constellation modeling; KAIST ground station visibility and feasible satellite analysis; Satellite link budget and data rate evaluation)
- Advanced English Listening; Advanced English Reading; (Summer, 2026 ; Prof : Josalyn Knapic)
- Reading Great Books on Human Intelligence and Civilization: Universe (Spring, 2026 ; Prof : Myung-Hyun Rhee)

Teaching Assistant

Korea University (KU) camp

- Taught students in drone assembly and flight practice, Arduino practice, and coding exercises (Summer, 2025)

Teaching Assistant

Korea Astronomy Olympiad (KAO)

- Summer 2023; Summer 2024; Winter 2024
- Instructed Astronomy Olympiad problems, taught astrodynamics supervised telescope observation

Social Service Agent (Military Duty) *Jan 2021 - Oct 2022*

Jongno Police Station

TEST SCORES

TOEFL *Dec 2025*

ETS (Educational Testing Service), United States of America *Score 96/120*

AWARDS & HONORS

Government-sponsored scholarship *Sep 2025*

KAIST

- Government-sponsored scholarship recipient during the Master's degree programme at KAIST

Full Scholarship

Yonsei University

- Granted full scholarship for the semester in recognition of outstanding academic performance (Fall, 2024)

Commendation Award (Certificate of Appreciation) *Oct 2022*

Jongno Police Station, Seoul, Korea

Full Scholarship

Sejong University

- Granted a full scholarship for the semester in recognition of first-rank academic performance (Spring, 2019; Spring, 2020)

VOLUNTEER WORK

Volunteer Teaching and Educational Mentoring,

- Over 250 hours of volunteer service
- Teaching mathematics to seniors at St. Ignatius Sogang University, a Korean GED night school (2025–present)
- Silver Lining, KAIST International Volunteer Club (Fall 2025–present)
- Taught mathematics to seniors at Sarangbang, a Korean GED night school (2021–2022)
- Provided mentoring, counseling, and mathematics instruction to a high school senior at an education welfare center (2021)

EXTRA CURRICULAR ACTIVITIES

SPACE Y

Public Astronomy Lecture *Jan 2024 - Present*

Yonsei University, SEOUL

- Delivered public astronomy lectures about satellite systems and astrodynamics for a general audience
- Organized an astronomical observation session

Speaking Society Club *Mar 2019 - Dec 2023*

Speaking Society Club (SSC), Hanyang University, Seoul

- Vice President, managed international members
- English conversation & debate club